CBCS SCHEME
SRI SAIRAM COLLEGE OF ENGINEERING

BCS508

USN | | | | | | | | | | |

Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Environmental Studies and E – Waste Management

Time: 1 hr. Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
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1. Which of the following conceptual spheres of the environment is having the least storage capacity of matter?
   a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere

2. In an eco-system, the flow of energy is
   a) Bidirectional b) Cyclic c) Unidirectional d) Multi Directional

3. Which of the following is not a prominent Chemical responsible for a good habitat?
   a) Oxygen b) Carbondioxide c) SO_2 d) Nutrients

4. In complex eco-systems, the degree of species Diversity is
   a) Poor b) Medium c) High d) None of these

5. World Environment day is on
   a) 5th May b) 5th June c) 18th July d) 14th August

6. Abiotic component includes
   a) Soil b) Temperature c) Water d) All of these

7. The sequence of eating and being eaten in an ecosystem is
   a) Food chain b) Carbon cycle c) Water cycle d) Anthropoco system

8. The organisms which directly feed on producer called
   a) Herbivores b) Carnivores c) Ominivores d) Decomposer

9. In an eco-system biological cycling of materials is maintains by
   a) Producer b) Consumer c) Decomposer d) All of these

10. Which pyramid is always upright
a) Energy b) Biomass c) Numbers d) Food chain

11. Blue baby syndrome is caused by contamination of water due to
a) Phosphates b) Sulphur c) Arsenic d) Nitrates

12. Major source of Flouride is
a) River water b) Ground water c) Food products d) Tooth paste

13. Out of the following Nutrients in Fertilizer which one causes minimum water pollution
a) Nitrogen b) Phosphorous c) Pottasium d) Organic matter

14. Excess fluorides in drinking water is likely to cause
a) Blue babies b) Fluorosis
c) Taste and odour d) Intestinal irritation

15. Disinfection of water is used to remove
a) Bacteria b) Odour c) Turbidity d) Colour

16. Brackish water is found in
a) Estuaries b) Salt lake c) Polar Ice caps d) Both a) and b)

17. Which one of the following is not a renewable energy?
a) Fossil Fuels b) Solar energy c) Wind energy d) Tidal wave energy

18. Natural gas consists mainly of
a) Ethane b) Propane c) Butane d) Methane

19. Which of the following is considered as an alternate promising Fuel?
a) CNG b) Kerosene c) Coal d) Petrol

20. Bio mass consists of
a) Lignin b) Hermicellulose c) Cellulose d) All of these

21. Nuclear power is being produced from
a) Carbon - 14 b) Uranium
c) Petroleum combustion d) Natural gas

22. Direct conversion of solar energy is attained by
a) Solar Photo voltaic system b) Solar diesel hybrid system
c) Solar thermal system d) Solar air heater

23. Pollutant waste water discharged from industries is known as
a) Effluent b) Sludge c) Pathogens d) Contaminant

24. Aanaerobiosis makes water
a) Clear b) Mudy c) Red d) Browinish or blackish

25. Which of the following is causes Entriphication
a) Nitrates b) Phosphates c) Heavy metals d) Both a) and b)

26. The word pollution is derived from the latin word
a) Pollute b) Pollutant c) Pollure d) Play- tant

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27. Which of the following are non-bio degradable?
a) Plastics b) Domestic sewage c) detergent d) a) and c)

28. Which of the following is secondary air pollutant?
a) Carbon monoxide b) Ozone c) Sulphur dioxide d) Carbon dioxide

29. Minamata Disease is caused due to
a) Lead b) Arsenic c) Mercury d) Cadmium

30. Increase in asthma attacks has been linked to high level of
a) Nitrogen b) Oxygen c) Air-borne particles d) All of these

31. Which of the following is an air pollutant?
a) Carbon b) Oxygen c) Nitrogen d) Particulate matter

32. Demography is the study of
a) Animals behaviour b) Population growth
c) River d) None of these

33. The urban solid waste is known as
a) Garbage b) Rubbish c) Refuge d) Silt

34. Activated sludge process is
a) Anaerobic method b) Aerobic method
c) Both a) and b) d) Land filling

35. Which toxic component not found in e-waste
a) Mercury b) Cadmium c) Neon d) Lead

36. Disposal of Bio-medical waste is by
a) Autoclave and land filling b) Incineration
c) Both a) and b) d) Landfilling

37. What does e-waste stands for
a) Environment waste b) Equipment waste
c) Electronic waste d) None of these

38. Which country producers maximum e-waste per year
a) India b) USA c) China d) France

39. What is the hazardous pollutant released in LED's?
a) Barium b) Arsenic c) Cobalt d) Cadmium

40. Under the E-Waste Management Rules 2016, who is responsible for the collection of e-waste from end-users
a) Retailer b) Manufacturers c) Bulk consumers d) Informal cycles

41. What percentage of e-waste is typically recycled Globally?
a) 10% b) 20% c) 40% d) 60%

42. What is the first step in the e-waste recycling process?
a) Shredding b) Sorting c) Melting d) Crushing

43. Which component often contain valuable metals like Gold, Silver and Copper?
a) Printed circuit boards b) Glass screens
c) Plastic casing d) Power cords

44. What are the health hazards which can caused by e-waste?
a) Lung cancer b) DNA Damage c) Brain d) All of these

45. Where should syringe Disposed?
a) Pharma pail b) Sharps container c) Gray tote d) Red pail

46. Which is the hazardous pollutant which can caused by e-waste, released from electron?
a) Arsenic b) Barium c) Cadmium d) Cobalt

47. Where should blood product pack to dispose?
a) Gray totte/yellow pail b) Regular waste
c) Pharma pail d) Red pail

48. Which one is Muncipal Solid Waste?
a) Toxic b) Hazardous c) Non-Toxic d) Non-Hazardous

49. Nickel is released from _____
a) Display b) Calculator c) Alloy d) None of these

50. Which is the hazardous pollutant occurs is plastic?
a) Lithium b) PCB's c) Lead d) Copper

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Environmental Studies and E – Waste Management

Time: 1 hr.] [Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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1. In an ecosystem biological cycling of materials is maintained by
   a) Producer b) Consumer c) Decomposer d) All of these

2. The primary producers in a forest ecosystem are
   a) Chlorophyll containing trees and plants b) Herbivores
   c) Carnivores d) Bacteria and other micro - organisms

3. Which of the following is the terrestrial ecosystem?
   a) Forest b) Grass land c) Desert d) All of these

4. Food web consists of
   a) a portion of food chain b) an organism position in a food chain
   c) interlocking food chain d) a set of similar consumers

5. The area of National Parks range between
   a) 0.61 to 7818 kms b) 0.04 to 3162 kms c) 0.14 to 3612 kms d) 0.16 to 8718 kms

6. Which Indian state has its maximum area under the forest cover?
   a) Maharashtra b) Madhya Pradesh c) Arunachal Pradesh d) Karnataka

7. How many parts are there in forest ecosystem?
   a) Biotic type b) Abiotic type c) Both (a) & (b) d) None of these

8. To which of the following is Ramsor convention related?
   a) Protection of wild life b) Protection of environment
   c) Protection of rivers d) Protection of wetlands

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9. Which of the following option is not included in the sustainable development parameters?
   a) Gender inequality
   b) Intergenerational and intragenerational equity
   c) Growing annually
   d) None of these

10. How many sustainable development goals are there?
   a) 10 b) 12 c) 13 d) 17

11. Which of the following are biodegradable pollutants?
   a) Plastics b) Domestic sewage c) Detergent d) All of these

12. BOD means
   a) Biochemical Oxygen Demand
   b) Chemical Oxygen Demand
   c) Biophysical oxygen Demand
   d) All of these

13. Sound beyond which of the following level can be regarded as a pollutant.
   a) 40 dB b) 80 dB c) 120 dB d) 150 dB

14. Which of the following is a point source of water pollution?
   a) Factories
   b) Sewage treatment plants
   c) Urban & Sub – urban lands
   d) Both (a) & (b)

15. Pesticide causes
   a) Eye irritation b) Skin irritation c) Respiratory ailment d) All of these

16. Amount of waste infectious produced in hospitals.
   a) 45 % b) 65 % c) 80 % d) 100 %

17. Cytotoxic and expired drugs are disposed of by
   a) Dumping b) Autoclave c) Incineration d) All of these

18. Color code of plastic bag for disposing of microbial laboratory culture waste __
   a) Black b) Red c) Blue d) White

19. All of the waste are incinerated except
   a) Reactive chemical waste
   b) Vaccine
   c) Mutilated parts
   d) Discarded drugs

20. The average composition of municipal solid waste is
   a) 41 % organic , 40 % inert & 19 % recyclable.
   b) 20 % organic , 60% inert & 20 % recyclable
   c) 30 % organic , 20% inert & 50 % recyclable
   d) 19 % organic , 41% inert & 40 % recyclable

21. Major atmospheric gas layer in stratosphere is
   a) Hydrogen b) Carbon dioxide c) Ozone d) Helium

22. World Environment day is on____
   a) 5th May b) 5th June c) 3rd Jan d) 1st June

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23. Which of the following is not a green house gas?
    a) Hydro chlorofluorocarbons
    b) Methane
    c) CO_2
    d) SO_2

24. Global warming could affect
    a) Climate
    b) Increase in sea level
    c) Melting of glaciers
    d) All of these

25. Primary cause of acid rain around the world is due to
    a) Carbon dioxide b) Sulphur dioxide c) Carbon monoxide d) Ozone

26. What is the main reason for the depletion of groundwater levels?
    a) Rains are less frequent
    b) Tree cover has reduced
    c) Tube wells and hand pumps are used for irrigation
    d) The course of river being changed and dams built

27. What happens when water contains 8 – 20 PPM of fluoride concentration?
    a) Blue baby disease
    b) Crippling fluorosis
    c) Dental fluorosis
    d) Mottling of teeth

28. Coral reefs of India are located in
    a) Goa
    b) Himalayan region
    c) Andaman & Nicobar Islands
    d) Uttar Pradesh

29. What is the allowable concentration of fluorides in drinking water?
    a) 1.0 mg/L b) 1.25 mg/L c) 1.5 mg/L d) 1.75 mg/L

30. Minmata disease is caused by
    a) Lead b) Mercury c) Cadmium d) Arsenic

31. When did National Disaster Management Authority form?
    a) 2000 b) 2005 c) 2010 d) 2015

32. Scientific study of earthquake is called
    a) Seismograph b) Seismology c) Both (a) & (b) d) None of these

33. Mass killing diseases can be referred as
    a) Biological disaster
    b) Industrial disaster
    c) War disaster
    d) Flood disaster

34. South Africa is leading exporter of which mineral?
    a) Copper b) Diamond c) Silver d) Gold

35. Which one of the following is an exhaustible resource?
    a) Coal b) Rain precipitation c) Air d) Tidal energy

36. Which of the following is a non – renewable resource?
    a) Solar energy b) Wind energy c) Fossil fuels d) Hydro power

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   c) 30 % organic , 20% inert & 50 % recyclable
   d) 19 % organic , 41% inert & 40 % recyclable

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   a) Hydrogen b) Carbon dioxide c) Ozone d) Helium

22. World Environment day is on____
   a) 5th May b) 5th June c) 3rd Jan d) 1st June

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37. Which of the following is an example of sustainable use of water resources?
a) Overuse of ground water
b) Building dams on rivers
c) Rainwater harvesting
d) Dumping industrial waste in rivers

38. Geothermal energy is a
a) Heat energy b) Current energy c) Wind energy d) Solar energy

39. Molasses from sugar industry is used to generate
a) Biodiesel b) Hydrogen c) Bio-ethanol d) Bio - methanol

40. Nuclear power plant in Karnataka is located at
a) Bhadravati b) Sandur c) Raichur d) Kaiga

41. Wild Life Protection Act in India was passed in
a) 1978 b) 1972 c) 1986 d) 1992

42. Environment Protection Act was enacted in year
a) 1986 b) 1974 c) 1992 d) 1984

43. Water Protection Act was enacted in
a) 1974 b) 1990 c) 1985 d) 2021

44. Blue baby syndrome is caused by the contamination of water due to
a) Phosphates b) Sulphur c) Arsenic d) Nitrates

45. Which toxic compound is not found in e – waste?
a) Mercury b) Cadmium c) Neon d) Lead

46. What are the health hazards caused by E – waste?
a) Lung cancer b) DNA damage c) Brain d) All of these

47. The Kyoto protocol was adopted at the
a) 3rd conference of UNFCC in 1977
b) Convention on the trans boundary effects of industrial accidents
c) UNFCC in 1992
d) Convention on Biological diversity

48. The primary cause of acid rain around the world is due to
a) Carbon dioxide b) Sulphur dioxide c) Carbon monoxide d) Ozone

49. The carbon "credit is permit" representing the right to emit.
a) One tonne of CO_2
b) 10 tonnes of CO_2
c) 5 tonnes of CO_2
d) 15 tonnes of CO_2

50. Fossil fuels are converted into energy by
a) Burning b) Cooling c) Sublimation d) Melting

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.] [Max. Marks: 50

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1. What is the Dissolved oxygen value required for the survival of aquatic species?
   a) 7 mg/L b) 8.2 mg/L c) 6.5 mg/L d) 4 mg/L

2. Which among the following is used to dump the waste collected in the cities?
   a) Land fills b) Ocean c) River d) All of these

3. Which type of waste includes items such as leftover food, fruit peels and yard trimmings?
   a) Hazardous waste b) Organic waste
   c) Bio – medical waste d) Electronic waste

4. Which of the integrated waste management is reduced on an individual level?
   a) Source Reduction b) Recycling c) Disposal d) Burning

5. What is called for the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions?
   a) Landfill b) Recycling c) Vermicomposting d) Incineration

6. The process of decomposition of biodegradable solid waste by earthworms is called
   a) Landfill b) Vermicomposting c) Composting d) Shredding

7. ____ is a liquid that passes through solid waste and extracts suspended impurities from it
   a) Leachate b) Sludge c) Distilled water d) Municipal

8. The colour code of plastic bag for disposing of microbial laboratory culture waste
   a) black b) red c) blue d) white

9. Average hospital waste produced per bed per day in Government hospital is
   a) 1.5 to 2 kg b) 0.5 – 4 kg c) 0.5 to 1 kg d) 0.5 – 2 kg

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10. Which of the following are the main contributors of the e –waste in the world?
    I. Refrigerators / freezers , washing machines , dishwashers.
    II. Small household appliances
    III. Personal computers, telephones , laptops , printers.
    IV. Gas cylinders, chimneys and home appliances
    a) Only I, II, III b) Only I & II c) Only I, III , IV d) All of these

11. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
    a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere

12. The ratio between energy flows at different points in a food chain is known as
    a) Ecological capacity b) Ecological efficiency
    c) Ecological assimilation d) Ecological potential

13. A predator is
    a) An animal that is fed upon another animal
    b) Animal that feeds upon both plants and animals
    c) An animal that feeds upon another animal
    d) A primary consumer

14. Why Rann of Kutch attracts aquatic birds in monsoon season?
    a) Because desert land is converted to forest land
    b) Because desert land is converted to snow
    c) Because desert land do not convert
    d) Because desert land is converted to salt marshes

15. Which kind of soil we can find on the surface of Thar desert?
    a) Rocky b) Moist c) Fertile d) Aeolian

16. Which of the following type of forest important for watersheds?
    a) Tropical Evergreen forests b) Tropical Deciduous forests
    c) Tropical Montana forests d) Grassland forest

17. Hot spots areas have
    a) Low density of biodiversity b) Only endangered plants
    c) High density of hot springs d) High density of biodiversity

18. Sustainable Development means
    a) meeting present needs without compromising on future needs
    b) progress of human beings
    c) balance between human needs and the ability of earth to provide the resources
    d) all of these

19. The term Alpha diversity refers to
    a) Genetic diversity b) Community and ecosystem diversity
    c) Species diversity within a community or ecosystem
    d) Diversity among the plant

20. Algae , green plants and photosynthetic bacteria are
    a) Autotrophic b) Heterotrophic c) Decomposers d) Consumers

21. What is Extended Producer Responsibility (EPR) as per the e – waste management rules in India?
    a) The responsibility of consumer to manage e – waste
    b) The responsibility of manufactures to manage e – waste throughout the product life cycle
    c) The responsibility of retailers to manage e – waste disposal
    d) The responsibility of informal recyclers to manage e – waste.

22. Which international agreement regulates the transboundary movements of hazardous waste, including e – waste?
    a) Kyoto Protocol b) Paris agreement
    c) Montreal Protocol d) Basel convention

23. Which colour bin is used for e – waste?
    a) Blue b) Green c) Yellow d) Black

24. What are the health hazards which can be caused by E – waste?
    a) Lung cancer b) DNA damage c) Brain d) All of these

25. Preparation of Guidelines for Environmentally sound Management of e – waste is a duty assigned to
    a) Producer b) Consumer c) MOEFCC d) SP CB/PCC

26. What is India's global rank in e –waste?
    a) 3 b) 13 c) 23 d) 33

27. When did the Karnataka State Pollution Control Board for prevention and control of water pollution constituted?
    a) 1974 b) 1978 c) 1982 d) 1986

28. Aerosol consisting of liquid droplets is called as
    a) Mist b) Dust c) Fog d) Acrosol

29. Which of the following is non – point source of water pollution?
    a) Factories b) Sewage treatment plant
    c) Urban and suburban land d) All of these

30. When is World Water day celebrated?
    a) January 26th b) June 5th c) September 22nd d) March 22nd

31. _____ is caused by drinking water high in nitrates.
    a) Cholera b) Kidney problem
    c) Liver problem d) Methomoglobinemia

32. Bhopal gas tragedy took place in the year ____ and the gas responsible was ____
    a) 1964, Hydrogen fluoride b) 1974, Methyl chloride
    c) 1984, methyl ISO – cyanide d) 1994, Methyl sulphate

33. The major chemical pollutants in photochemical smog are
    a) NO, NO_2 , VOC , O_3 , PAN b) N_2O , NO_2 , VOC, O_3 , PAN
    c) NO, NO_2 , VOC , O_2 , PAN d) NO, N_2O_5, VOC, O_3 , PAN

34. The international protocol to protect the Ozone layer is
a) Vienna protocol b) Kyoto protocol
c) Cartagena protocol d) Montreal protocol

35. Which is the best and the worst method of plume behavior for pollution dispersion?
a) Lofting and fumigation b) Trapping and fanning
c) Conning and fumigation d) Fanning and Lofting

36. What is called for a Temporary hearing loss?
a) Temporary ear pain b) Temporary hearing problem
c) Temporary threshold shift d) Temporary hearing shift

37. What timings loud speakers shouldn't use in public areas?
a) 10.00 pm to 5.00 am b) 11.00 pm to 6.00 am
c) 1.00 am to 7.00 am d) 10.00 pm to 6.00 am

38. In which section, if a person violates the noise pollution regulations , is liable for penalty according to Environmental Protection Act, 1986.
a) Section 12 b) Section 15 c) Section 18 d) Section 19

39. 'Minamata Disease" is caused due to
a) Lead b) Arsenic c) Mercury d) Cadmium

40. The process of reducing the fluoride content from water is called as
a) Chlorination b) Fluoridation
c) Defluoridation d) Fanning and Lofting

41. Veld type grasslands are located at
a) South Africa b) South America c) Australia d) Britain

42. Which pyramid is always upright?
a) Energy b) Biomass c) Numbers d) Food chain

43. In what form is solar energy is radiated from the Sun?
a) Ultraviolet Radiation b) Infrared Radiation
c) Electromagnetic waves d) Transverse waves

44. What does MHD stands for in the energy field?
a) Magneto Hydro Dynamic b) Metal Hydrogen Detox
c) Micro Hybrid Drive d) Metering Head Differential

45. The 'Miracle Material' that can turn CO_2 into liquid fuel is :
a) Propane b) Copper c) Graphene d) Potassium

46. A tide whose difference between high and low tide is greatest.
a) Diurnal tide b) Neap tide c) Spring tide d) Ebb tide

47. Which of the turbine can be mounted vertically and horizontally.
a) Pelton wheel b) Kaplan turbine c) Gorlov turbine d) Francis turbine

48. Which type of fuel is removed from the reactor core after reaching end of core life service?
a) Burnt fuel b) Spent fuel c) Engine oil d) Radioactive fuel

49. What is a fuel cell?
a) Converts heat energy to chemical energy
b) Converts heat energy to electrical energy
c) Converts chemical energy to electrical energy
d) Converts kinetic energy to heat energy

50. Which one of the following is the apex organization in our country in the field of pollution control?
a) Water Pollution Control Board b) State Pollution Control Board
c) Central Pollution Control Board d) Air Pollution Control Board